

一、解:

特征值 $\lambda_1 = \lambda_2 = 0, \lambda_3 = -1$

$$\text{对 } \lambda = 0, A - 0E = \begin{bmatrix} -1 & 1 & 1 \\ -5 & 21 & 17 \\ 6 & -26 & -21 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & -1 \\ 0 & 16 & 12 \\ 0 & -20 & -15 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & -1 \\ 0 & 4 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{rank}(A) = 2, q_1 = n - \text{rank}(A) = 1$$

$$\text{对 } \lambda = -1, A + E = \begin{bmatrix} 0 & 1 & 1 \\ -5 & 22 & 17 \\ 6 & -26 & -20 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

→ 一次幂不用求秩, 对应一个 Jordan 块

$$\text{rank}(A+E) = 2, q_2 = n - \text{rank}(A+E) = 1$$

$$\therefore J = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

由 $P^{-1}AP = J$ 得 $AP = PJ$, 设 $P = [x_1, x_2, x_3]$

$$\text{则 } AP = [Ax_1, Ax_2, Ax_3] = PJ = [-x_1, 0, x_2]$$

$$\therefore \begin{cases} Ax_1 = -x_1 \\ Ax_2 = 0 \\ Ax_3 = x_2 \end{cases} \quad \text{即: } \begin{cases} (A+E)x_1 = 0 \\ Ax_2 = 0 \\ Ax_3 = x_2 \end{cases}$$

$$\text{求 } (A+E)x_1 = 0 \text{ 得: } \alpha_1 = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$$

$$\text{求 } Ax_2 = 0 \text{ 得: } \alpha_2 = \begin{bmatrix} 1 \\ -3 \\ 4 \end{bmatrix} \quad \text{故 } P = [x_1, x_2, x_3] = \begin{bmatrix} -1 & 1 & -5 \\ -1 & -3 & -5 \\ 1 & 4 & 4 \end{bmatrix}$$

$$\text{求 } Ax_3 = x_2 \text{ 得: } \alpha_3 = \begin{bmatrix} -5 \\ -5 \\ 4 \end{bmatrix}$$

二、解:

$$AA^H = \begin{bmatrix} 8 & 4 & 0 \\ 4 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$|AA^H - \lambda E| = \lambda^2(\lambda - 10), \lambda_1 = \lambda_2 = 0, \lambda_3 = 10$$

$$\text{对 } \lambda = 0, AA^H - 0 \cdot E = \begin{bmatrix} 8 & 4 & 0 \\ 4 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 2 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \alpha_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \alpha_2 = \begin{bmatrix} -\frac{1}{2} \\ 1 \\ 0 \end{bmatrix}$$

$$\text{单位正交化 } \beta_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \beta_2 = \begin{bmatrix} -\frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \\ 0 \end{bmatrix}$$

$$\text{对 } \lambda = 10, AA^H - 10E = \begin{bmatrix} -2 & 4 & 0 \\ 4 & -8 & 0 \\ 0 & 0 & -10 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \alpha_3 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \text{ 单位化 } \beta_3 = \begin{bmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \\ 0 \end{bmatrix}$$

$$U = \beta_3, U = \begin{bmatrix} \frac{2}{\sqrt{5}} & 0 & -\frac{1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} & 0 & \frac{2}{\sqrt{5}} \\ 0 & 1 & 0 \end{bmatrix}, V_1 = A^H U \Delta^{-1} = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \\ 0 \end{bmatrix}, V = \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ 0 & 0 \end{bmatrix}$$

$$\text{则 } A = U \begin{bmatrix} 10 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} V^H$$

三. 解

$$|A - \lambda E| = \begin{vmatrix} 2-\lambda & 1 & 0 \\ 1 & 2-\lambda & 0 \\ -2 & 2 & 3-\lambda \end{vmatrix} = (3-\lambda) \begin{vmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{vmatrix} = -(\lambda-1)(\lambda-3)^2$$

$$\therefore \lambda_1 = 1, \lambda_2 = \lambda_3 = 3$$

$$A - 3E = \begin{bmatrix} -1 & 1 & 0 \\ 1 & -1 & 0 \\ -2 & 2 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad q = n - \text{rank}(A - 3E) = 2$$

$$\therefore A \text{ 是对角化矩阵 且 } \alpha_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad \alpha_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$A - E = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ -2 & 2 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \therefore \alpha_3 = \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 1 \end{bmatrix}$$

$$\therefore P = [\alpha_1 \alpha_2 \alpha_3] = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & -1 \\ 1 & 0 & 2 \end{bmatrix}, \quad P^{-1} = \begin{bmatrix} -1 & 1 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & -\frac{1}{2} & 0 \end{bmatrix} = \begin{bmatrix} \beta_1^T \\ \beta_2^T \\ \beta_3^T \end{bmatrix}$$

$$\text{令 } G_1 = \alpha_1 \beta_1^T + \alpha_2 \beta_2^T = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ -1 & 1 & 1 \end{bmatrix} \quad G_2 = \alpha_3 \beta_3^T = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & 0 \\ -\frac{1}{2} & \frac{1}{2} & 0 \\ 1 & -1 & 0 \end{bmatrix}$$

其谱分解表达式为 $\dot{A} = 3G_1 + G_2$

四. 解: 考虑幂级数 $\sum_{k=0}^{\infty} (2k+1)x^k$

$$\begin{aligned} \sum_{k=0}^{\infty} (2k+1)x^k &= \sum_{k=0}^{\infty} (2kx^k + x^k) = 2 \sum_{k=0}^{\infty} kx^k + \sum_{k=0}^{\infty} x^k \\ &= 2 \left(\sum_{k=0}^{\infty} x^{k+1} \right)' + \sum_{k=0}^{\infty} x^k = 2 \cdot \left(\frac{x^2}{1-x} \right)' + \frac{x}{1-x} \quad |x| < 1 \\ &= \frac{3x - x^2}{(1-x)^2}, \quad R=1 \end{aligned}$$

$$(1) |A - \lambda E| = \begin{vmatrix} \frac{1}{2} - \lambda & -a \\ -a & \frac{1}{2} - \lambda \end{vmatrix} = (\frac{1}{2} - \lambda)^2 - a^2 = 0 \Rightarrow \lambda = \frac{1}{2} \pm a$$

$$\rho(A) = \max(|\lambda|) = \max(|\frac{1}{2} + a|, |\frac{1}{2} - a|)$$

要使矩阵幂级数绝对收敛, 应使 $\rho(A) < 1$

$$\text{若 } a \geq 0 \text{ 则 } \rho(A) = \frac{1}{2} + a \quad \therefore a < \frac{1}{2}$$

$$\text{若 } a < 0 \text{ 则 } \rho(A) = \frac{1}{2} - a \quad \therefore a > -\frac{1}{2}$$

综上, a 的范围是 $-\frac{1}{2} < a < 0$ 或 $0 \leq a < \frac{1}{2}$

$$(2) \text{ 当 } a=0 \text{ 时, } A = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{bmatrix}, \quad A^2 = \begin{bmatrix} \frac{1}{4} & 0 \\ 0 & \frac{1}{4} \end{bmatrix}$$

$$\sum_{k=0}^{\infty} (2k+1)A^k = \frac{3A - A^2}{(E - A)^2} = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$$

五. 解: $|A - \lambda E| = -(\lambda-2)^2(\lambda-4)$

$$A - 2E = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad r=1$$

$$\text{故 Jordan 标准形为 } \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$$Q_2(\lambda) = (\lambda-2)(\lambda-4)$$

$$\text{设 } p(x) = a_0 + a_1 x$$

$$p(2) = f(2) = a_0 + 2a_1$$

$$p(4) = f(4) = a_0 + 4a_1 \Rightarrow \begin{cases} a_0 = 2f(2) - f(4) \\ a_1 = \frac{1}{2}[f(4) - f(2)] \end{cases}$$

$$\text{故 } f(A) = [2f(2) - f(4)]E + \frac{1}{2}[f(4) - f(2)]A$$

$$e^{At}: f(x) = e^{tx}, f(2) = e^{2t}, f(4) = e^{4t}$$

$$f(A) = (2e^{2t} - e^{4t})E + \frac{1}{2}(e^{4t} - e^{2t})A$$

$$= \begin{bmatrix} \frac{1}{2}e^{4t} + \frac{1}{2}e^{2t} & 0 & 1 \\ 1 & e^{2t} & 1 \\ 1 & 0 & \frac{1}{2}e^{2t} + \frac{1}{2}e^{4t} \end{bmatrix}$$

$$\sin \pi A: f(x) = \sin \pi x, f(2) = 0, f(4) = 0$$

$$f(A) = 0$$

八. 证明:

设 $A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$, $x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$

则 $Ax = \begin{bmatrix} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n \end{bmatrix}$

$$\begin{aligned} \text{令 } f(x) &= x^H (A^H A) x \\ &= x^H A^H A x = (Ax)^H (Ax) \\ &= (a_{11}x_1 + \cdots + a_{1n}x_n)^2 + \cdots + (a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n)^2 \geq 0 \end{aligned}$$

当且仅当 $\begin{cases} a_{11}x_1 + \cdots + a_{1n}x_n = 0 \\ a_{21}x_1 + \cdots + a_{2n}x_n = 0 \\ \vdots \\ a_{n1}x_1 + \cdots + a_{nn}x_n = 0 \end{cases}$ 时, $f(x) = 0$

即: $Ax = 0$ 时 $f(x) = 0$

又 $R(A) = n$ $\therefore Ax = 0$ 只有零解 $x = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$ $\therefore f(x)$ 为正定二次型

$A^H A$ 为正定矩阵 又 $(A^H A)^H = A^H A$ $\therefore A^H A$ 为 Hermite 矩阵. 得证.

七. 证: $Ax = b$

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 2 & 3 & 3 \end{bmatrix}, b = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$$

$$A = BC = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

$$A^+ = C^H (CC^H)^{-1} (B^H B)^{-1} B^H$$

$$x = A^+ b$$

九. 证:

(1) $\varphi_{J_0}(\lambda) = (\lambda - \lambda_0)^m$

(2) 由 $P^{-1}AP = J$ 得: $P^{-1}f(A)P = f(J)$

$\therefore \varphi_A(\lambda) = \varphi_J(\lambda)$

(3) A 与 J 有相同的最小多项式

故 J 的最小多项式为 $\varphi_J(\lambda) = (\lambda - \lambda_1)(\lambda - \lambda_2) \cdots (\lambda - \lambda_k)$

因此 $P^{-1}AP = J = \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_k \end{bmatrix}$

A 可对角化, 即是单纯矩阵.